Assignment 1

Underactuated Robotics and Manipulation

Acrobots, Cart-Poles, Quadrotors, Walking Models,
Legged Robots, and an Introduction to Manipulation

Zihao Liu
Student ID: 220261979
213222647@seu.edu.cn

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Abstract

This report documents the four parts of Assignment 1: three chapters of *Underactuated Robotics* (acrobots, cart-poles and quadrotors; simple models of walking and running; highly-articulated legged robots) and the introduction of *Robotic Manipulation*. I ran the supplied notebooks, reproduced their main results, and added parameter and robustness studies where they were useful. The visible autograders pass: 13/13 and 12/12 for the cart-pole notebooks, 13/13 for the hopper, 16/16 for the two footstep planners, and 70/70, 50/50 and 50/50 for the manipulation exercises. The report gives the steps behind these numbers, the plots that show the behaviour, and the code that produced them. The less obvious cases—saturation, non-minimum-phase zeros, the narrow compass-gait basin, and the partial stability of the SLIP apex map—are discussed with the numerical evidence rather than only stated as conclusions.

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Chapter 1

Introduction

1.1 Scope of the assignment

Assignment 1 covers four textbook chapters. The first three belong to *Underactuated Robotics*; the last one is the introduction of *Robotic Manipulation*. The required examples and exercise sections are summarised in Table 1.1.

Table 1.1: Assignment 1 scope.

Textbook chapter	Examples	Exercise section
Ch. 3 Acrobots, Cart-Poles, and Quadrotors	3.1, 3.2, 3.5, 3.6	Section 3.8, Exercises 3.1–3.4
Ch. 4 Simple Models of Walking and Running	4.3, 4.4, 4.5, 4.6	Section 4.6, one-dimensional hopper
Ch. 5 Highly-articulated Legged Robots	5.1, 5.2	Section 5.8, three exercises
Ch. 1 Introduction (Manipulation)	1.2, 1.3	Section 1.7, Exercises 1.1–1.4

The examples are the supplied demonstrations: the code is already written and the task is to run it, explain what is observed, and, where possible, study the effect of changing a parameter. The exercise sections are solved from scratch; the report includes the derivations, the important code, and the relevant plots.

1.2 Environment and code

All computations use Drake 1.58.0, the `underactuated` package 2026.9.8, and the `manipulation` package 2026.10.17 in the Python environment `/Users/leo/venvs/drake/bin/python`. The notebooks were executed headlessly with `COLAB_TESTING=1`; this prevents the MeshCat visualizer from blocking on a browser connection. The pinned `drake_models` package is used for the IIWA14, the WSG gripper and the YCB object in the manipulation chapter.

Table 1.2: Coverage of the assignment items and where the corresponding code and results can be found.

Assignment item	Section	Code / notebook	Main result
Example 3.1	§2.1	`exp1_analysis.py`, `Exp3'1-acrobot.ipynb`	98% / 63%
Example 3.2	§2.2	`cp_analysis.py`, `Exp3'2-cartpole.ipynb`	100% / 36%
Example 3.5	§2.3	`lqr_comparison.py`	conditions, zeros, LQR
Example 3.6	§2.4	`quadrotor_analysis.py`	settles in 1.08 s
Section 3.8	§2.5	`cartpole_balancing.ipynb`, `cartpoles_urdf.ipynb`	13/13, 12/12
Example 4.3	§3.1	`rimless_wheel_analysis.py`	fixed point $\omega^\star$
Example 4.4	§3.2	`compass_gait_analysis.py`	14 steps / 0.735 s
Examples 4.5–4.6	§3.3	`slip_analysis.py`	$z^\star = 0.8776$ m
Section 4.6	§3.4	`one_d_hopper_analysis.py`	13/13
Example 5.1	§4.1	`zmp_analysis.py`	0.153 m
Example 5.2	§4.2	`littledog_analysis.py`	0.6875 s stride
Section 5.8	§4.3	`footstep_analysis.py`	MIQP/GCS 16/16
Example 1.2	§5.1	teleoperation notebook	3D demo ran
Example 1.3	§5.2	this report	block diagrams
Section 1.7	§5.3	four notebooks	70/50/50

The complete code accompanies the report as `Assignment_1_Code.zip`. The files are listed in Appendix A. For the reader who wants to reproduce a specific result, the appendix gives the exact commands.

1.3 Summary of results

Table 1.3 collects the automatic grading results. They are useful as a compact check that the code matches the expected behaviour; the discussion in the following chapters explains what the numbers mean.

Table 1.3: Automatic grading results.

Chapter	Item	Result
Ch. 3	`cartpole_balancing.ipynb`	13/13
Ch. 3	`cartpoles_urdf.ipynb`	12/12
Ch. 4	`one_d_hopper.ipynb`	13/13
Ch. 5	MIQP footstep planning	16/16
Ch. 5	GCS footstep planning	16/16
Ch. 1	Drake Systems Fundamentals	70/70
Ch. 1	Physics Simulation and Robot Loading	50/50
Ch. 1	HardwareStation and Advanced Scenarios	50/50

Part I

Underactuated Systems

Chapter 2

Acrobots, Cart-Poles, and Quadrotors

2.1 Example 3.1: Dynamics of the Acrobot

The acrobot is a two-link planar robot whose shoulder is passive and whose elbow is driven by a torque $u = \tau$ [1]. With $q = [\theta_1, \theta_2]^\top$ and $\mathbf{x} = [\theta_1, \theta_2, \dot\theta_1, \dot\theta_2]^\top$, the manipulator equations are

$$M(q)\ddot q + C(q,\dot q)\dot q + G(q) = \begin{bmatrix} 0 \\ u \end{bmatrix}.$$

The upright equilibrium is $\mathbf{x}^\star = [\pi, 0, 0, 0]^\top$. At this point the linearisation obtained in the notebook is

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 12.629 & -12.693 & -0.172 & 0.3015 \\ -14.749 & 29.612 & 0.3015 & -0.6033 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ -3.0147 \\ 6.0332 \end{bmatrix},$$

with open-loop eigenvalues $\{-6.492, -2.249, +2.231, +5.734\}$. Two positive eigenvalues confirm that the upright position is unstable. The controllability matrix $\mathcal{C} = [\mathbf{B}, \mathbf{AB}, \mathbf{A}^2\mathbf{B}, \mathbf{A}^3\mathbf{B}]$ has rank four, so the linearised system is controllable even though the robot is underactuated. The singular values of $\mathcal{C}$ are $[511.7, 123.1, 0.0807, 0.0715]$; the ratio between the largest and the smallest is about 7×10^3, which indicates that some directions are only weakly controllable. The rank is four, but the smallest singular value is only $1.4 imes 10^{-4}$ of the largest. In the simulation this appeared as a large initial elbow motion; the shoulder moved only after the elbow had built up momentum.

With $Q = \mathrm{diag}(10, 10, 1, 1)$ and $R = 1$, the LQR gain is

$$D = [\,278.44,\ 112.29,\ 119.72,\ 56.83\,], \qquad u = D(\mathbf{x} - \mathbf{x}^\star),$$

and the closed-loop poles are -10.126, -4.238 and $-2.242 \pm 0.039j$. A 100-trial study with initial state $\mathbf{x}_0 = \mathbf{x}^\star + 0.05\,\mathcal{N}(0, I_4)$ gives a 98% success rate without saturation and 63% with the ± 10 N m actuator limit. The successful trajectories show a pronounced non-minimum-phase undershoot: the elbow first moves away from the target before the whole robot returns.

Two parameters are especially important. Increasing the distal mass m_2 from 0.5 to 4 kg reduces the minimum controllability singular value by a factor of about 35 and the success rate from 98% to 10%; the peak torque grows from 5 N m to more than 300 N m. Changing R from 1 to 100 reduces the peak torque only from 11.2 to 9.7 N m, because the unstable open-loop poles impose a finite gain even in the expensive-control limit. In practice, the acrobot needs a swing-up phase to enter the LQR region, and the actuator limit must be included in the assessment.

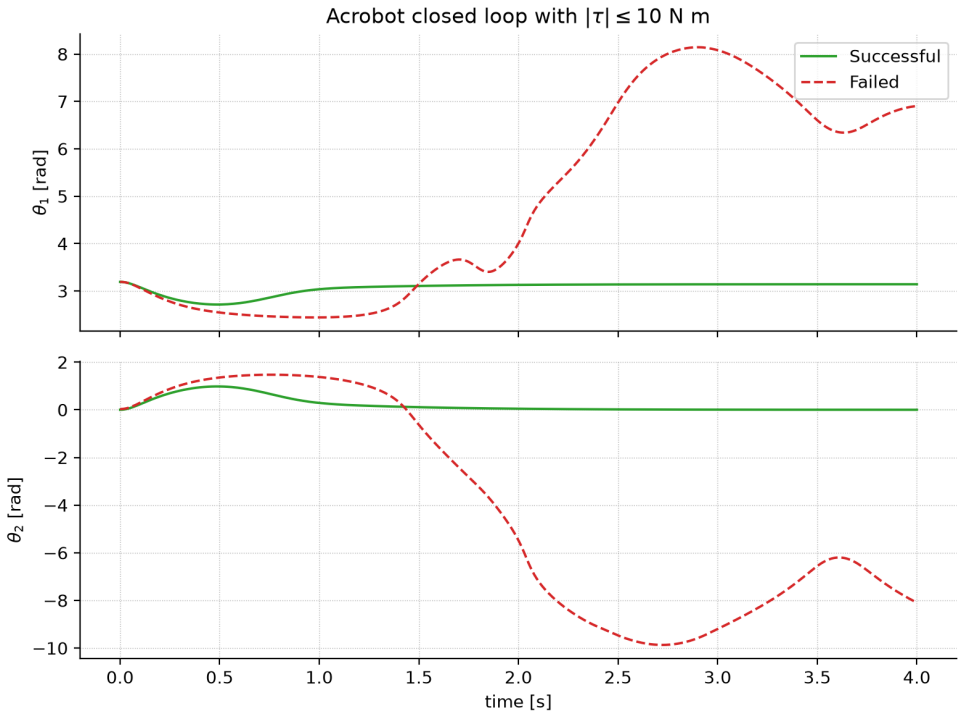


Figure 2.1: Acrobot with the ± 10 N m saturation. The green run starts with a 0.05 rad shoulder error; the red run also has a 0.03 rad elbow error and saturates. The elbow moves away from zero before the shoulder returns, which is the non-minimum-phase undershoot.

2.2 Example 3.2: Dynamics of the Cart-Pole

The cart-pole has state $\mathbf{x} = [x, \theta, \dot{x}, \dot{\theta}]^\top$ and input $u = F_x$. For the model used in the notebook, $m_c = 10$ kg, $m_p = 1$ kg and $l_c = 0.5$ m, the linearisation at $\mathbf{x}^\star = [0, \pi, 0, 0]^\top$ is

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0.981 & 0 & 0 \\ 0 & 21.582 & 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 0.1 \\ 0.2 \end{bmatrix}.$$

The open-loop eigenvalues are $0, 0, \pm 4.6456$. The two zero eigenvalues come from the cart position, which behaves as an integrator; the pair ± 4.6456 is the unstable pole. The controllability matrix is full rank, with singular values $[4.326, 4.326, 0.0907, 0.0907]$. The transfer function from u to x has a right-half-plane zero at $+4.43$, so the cart must first move in the opposite direction before it can bring the pole back. This is the same non-minimum-phase mechanism as in the acrobot, seen more clearly in the cart position.

With $Q = \mathrm{diag}(10, 10, 1, 1)$ and $R = 1$,

$$D = [\,3.1623,\ -254.3925,\ 9.7692,\ -55.1774\,],$$

and the closed-loop poles are -4.7206, -4.5739 and $-0.3820 \pm 0.3761j$. The two fast poles bring the pole back; the slow pair is the cart returning to the origin. The original notebook does not limit the force, and a 100-trial study with $\mathbf{x}_0 = \mathbf{x}^\star + 0.1\,\mathcal{N}(0, I_4)$ balances the pole in 100% of the trials. The median peak force is about 17 N, with a 90th percentile near 40 N. When the force is limited,

the picture changes: 20 N gives 90% pole-balancing success, 5 N gives 36%, and 2 N fails in most trials because the non-minimum-phase recovery needs a larger control authority.

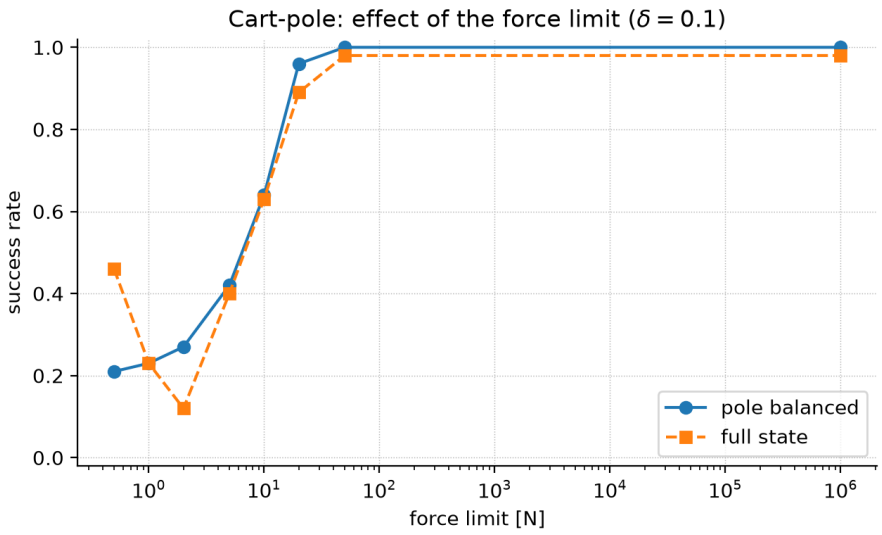


Figure 2.2: Cart-pole success rate versus force limit at $\delta = 0.1$. At 20 N the pole still balances in 90% of trials; at 10 N the rate is 64%, and at 5 N only 36%. Without a limit the pole is easy; the actuator is the binding constraint.

The cart mass is the most important physical parameter. From 2 to 20 kg the median peak force grows from about 5 to 32 N, and the full-state success rate falls from 100% to 68%; the pole itself remains balanceable. Increasing R saves force but slows the cart-return mode: for $R = 100$ the slow pole is about -0.12 rad/s, so a 5 s simulation is too short for the cart to return.

2.3 Example 3.5: LQR for the Acrobot and Cart-Pole

The acrobot and the cart-pole are useful together because they have the same control structure but different modal properties. Both are controllable and both are non-minimum phase. The acrobot's condition number is about 150 times larger, which reflects the weaker coupling through the passive shoulder. The cart-pole has two zero eigenvalues and therefore a slow cart mode, while the acrobot has four relatively fast modes. These differences explain why the same LQR weight has a different effect on the two systems: increasing R mainly reduces the torque on the acrobot, but it substantially slows the cart return on the cart-pole.

2.4 Example 3.6: LQR for Quadrotors

The planar quadrotor has state $\mathbf{x} = [x, z, \theta, \dot{x}, \dot{z}, \dot{\theta}]^\top$ and two thrust inputs. At hover, $u_1 + u_2 = mg$ and $\theta = 0$. The textbook LQR uses

$$\mathbf{Q} = \operatorname{diag}\left(10, 10, 10, 1, 1, \frac{l}{2\pi}\right), \qquad \mathbf{R} = \begin{bmatrix} 0.1 & 0.05 \\ 0.05 & 0.1 \end{bmatrix}.$$

The closed-loop poles are -80.75, -15.86, -6.59, -3.60 and $-2.72 \pm 1.61j$. From an initial tilt of 0.2 rad the controller settles in about 0.06 s with a peak thrust of 5.17 N; ten random initial conditions with standard deviation 0.1 all recover.

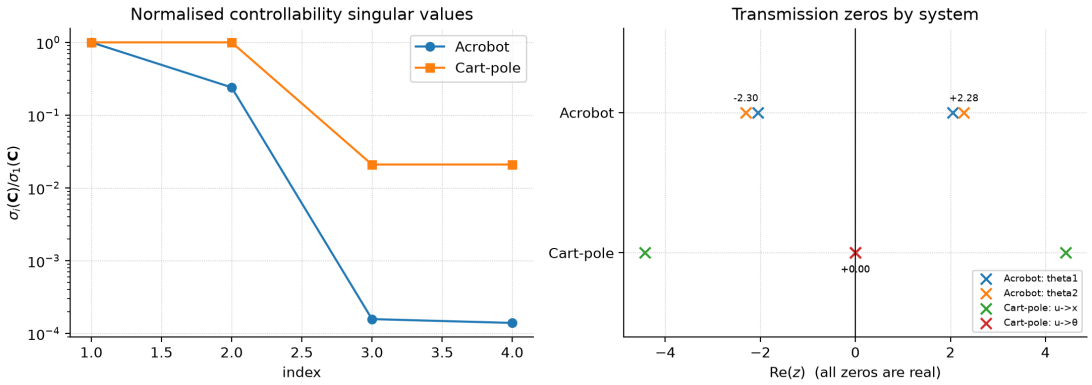


Figure 2.3: Normalised controllability singular values and transmission zeros. The acrobot's smallest normalized singular value is about 1.4×10^{-4}; the cart-pole's is about 2.1×10^{-2}. Both have right-half-plane zeros: the acrobot at $+2.05$ and $+2.28$, the cart-pole at $+4.43$.

The sensitivity study shows the expected trade-offs. Increasing the position weight makes the response faster but raises the peak thrust. Increasing R reduces the peak thrust at the cost of a slower response. A constant wind produces a steady-state offset because LQR has no integral action: the offset is about 0.073 m at 0.25 N and 0.554 m at 2 N. The initial tilt is recovered for all tested values up to 0.7 rad.

The 3D quadrotor has four thrust inputs and six degrees of freedom. It is underactuated, but the hover equilibrium is stabilisable. The LQR is linearised around a point where the total thrust balances gravity, not around zero thrust. From a 0.2 rad roll and 0.1 rad pitch perturbation the official stabilising controller and our LQR both return to hover in about 1.08 s with a peak thrust near 3 N. The sensitivity to Q, R and initial tilt follows the same pattern as in the planar case.

2.5 Section 3.8: Exercises 3.1–3.4

2.5.1 Exercise 3.1: linearisation and balancing

For the cart-pole with $m_c = m_p = l = 1$ and $g = 9.81$, the equations used in the notebook are

$$\ddot{x} = \frac{u + \sin\theta(\dot{\theta}^2 + g\cos\theta)}{1 + \sin^2\theta}, \qquad \ddot{\theta} = \frac{-u\cos\theta - \dot{\theta}^2\cos\theta\sin\theta - 2g\sin\theta}{1 + \sin^2\theta}.$$

At the upright point $\mathbf{x}^\star = [0, \pi, 0, 0]^\top$,

$$\mathbf{A}_{\text{lin}} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 9.81 & 0 & 0 \\ 0 & 19.62 & 0 & 0 \end{bmatrix}, \qquad \mathbf{B}_{\text{lin}} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

The linearisation error is small near the equilibrium and grows quickly with distance. At $\theta = 0.99\pi$ the error is 9×10^{-4}; at $\theta = \pi/2$ it is 26.05; at $\theta = 0$ it is 68.91. The error also depends on the input when $\theta \neq \pi$, because the nonlinear input gains $1/(1 + \sin^2\theta)$ and $-\cos\theta/(1 + \sin^2\theta)$ are replaced by constants in the linear model. The cart position x does not appear in the dynamics, so it does not affect the error.

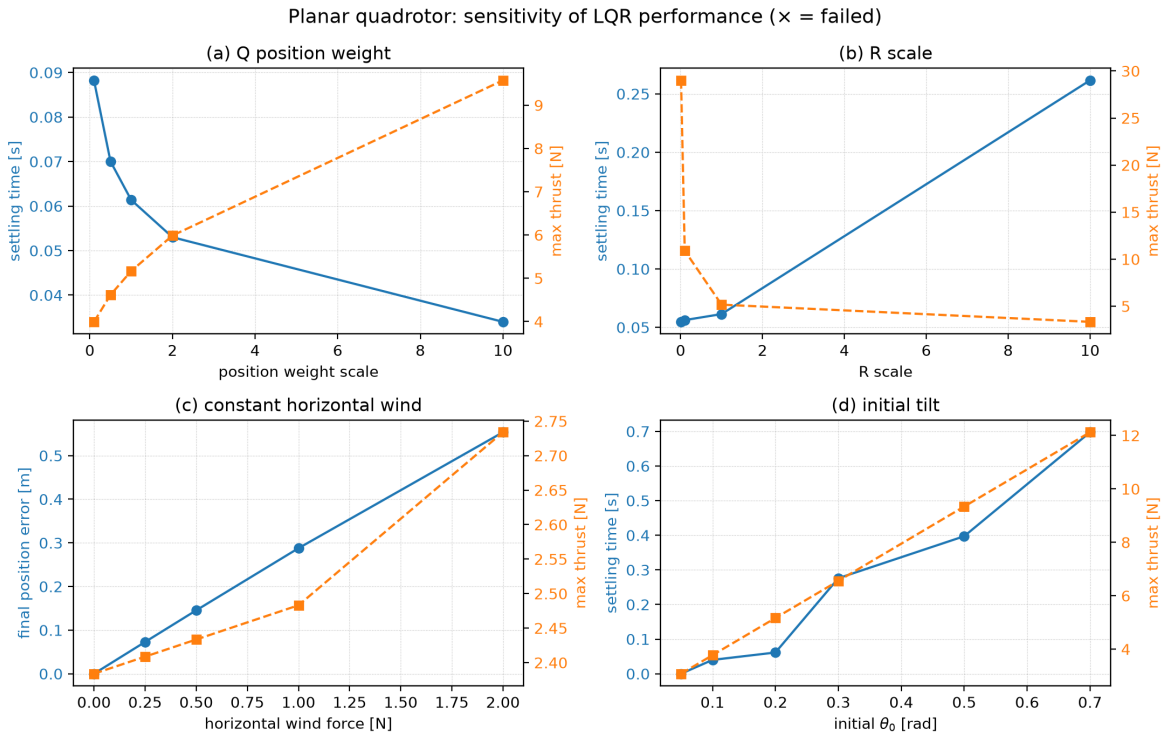


Figure 2.4: Planar quadrotor sensitivity. With 0.25 N wind the steady-state offset is 0.073 m; with 2 N it is 0.554 m. The other panels show the expected Q/R trade-off.

With $Q = I_4$ and $R = 1$, the LQR gain is

$$K = [-1.0000,\ 51.0612,\ -2.7221,\ 12.8277], \qquad u = -K(\mathbf{x} - \mathbf{x}^\star),$$

and the closed-loop poles are $-5.0417, -3.9438, -0.5600 \pm 0.4239j$. The six initial states from the notebook are simulated for 5 s; the controller recovers from states 0, 1, 2 and 5:

$$\texttt{system_recovers_from_states} = [0, 1, 2, 5].$$

The first three are close to the upright point. State 5 has $\theta = \pi$ and only a cart offset, which is linearly controllable. States 3 ($\theta = \pi/2$) and 4 ($\theta = 0$) lie outside the region where the linearisation is useful; the controller fails there, as the error table predicts.

2.5.2 Exercise 3.2: cart-pole URDFs

The single pendulum URDF has a base of mass 1 at the origin, a pendulum whose ball of mass 1 is at $z = -1$, a spherical visual of radius 0.05 and a cylindrical rod of length 1 and radius 0.01 centred at $z = -0.5$. The joint `theta0` connects the base to the pendulum, with origin $(0, 0, 0)$ and axis $(0, -1, 0)$. The unstable equilibrium is $\mathbf{x}^\star = [0, \pi, 0, 0]^\top$.

The double pendulum needs more care with the frames. The second joint is at the ball of the first pendulum, i.e. at $(0, 0, -1)$ in the `pendulum0` frame. The second link has its ball at $z = +1$ in its own frame. With $\theta_1 = 0$ the second pendulum points back towards the cart and overlaps the first; with $\theta_0 = \theta_1 = \pi$ both pendula stand upright. The equilibrium is $\mathbf{x}^\star = [0, \pi, \pi, 0, 0, 0]^\top$.

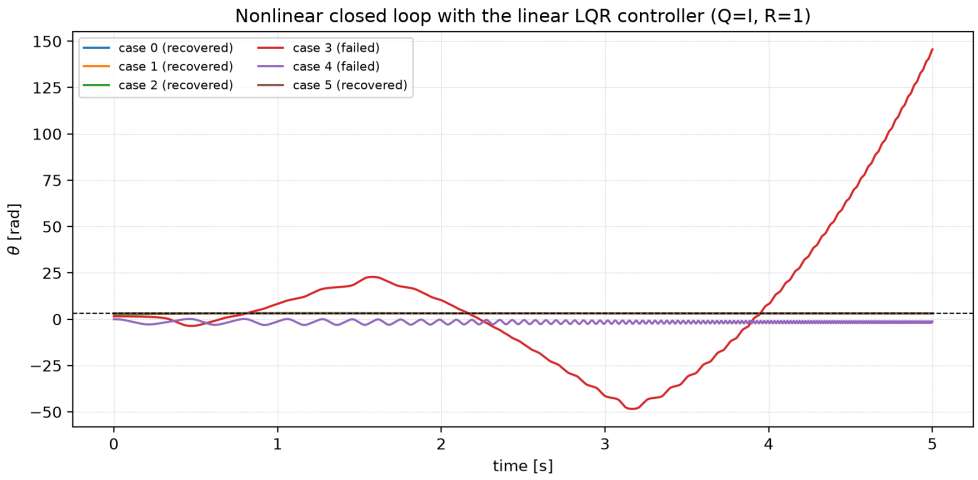


Figure 2.5: Closed-loop angle histories for the six initial states of Exercise 3.1. Cases 3 ($\theta = \pi/2$) and 4 ($\theta = 0$) leave the upright region; the linear controller was designed for the tangent at π.

The LQR is designed directly from the nonlinear `MultibodyPlant`. For the single pendulum with $Q = \mathrm{diag}(10, 10, 1, 1)$ and $R = 1$,

$$K = [-3.1623,\ 61.3987,\ -5.2854,\ 16.1699],$$

with poles $-4.8948, -4.0983, -0.9457 \pm 0.8075j$. After 5 s the angle error is 0.0036 rad and the peak force is 19.3 N. For the double pendulum with $Q = \mathrm{diag}(10, 10, 10, 1, 1, 1)$,

$$K = [3.1623,\ -54.8416,\ -300.4370,\ 7.4384,\ -54.0081,\ -68.9418],$$

with poles $-7.5798, -6.1385, -3.6749, -3.4281, -0.7754 \pm 0.6611j$. The two angle errors after 5 s are 0.0142 and 0.0007 rad, and the peak force is 66.6 N. The autograder gives 12/12.

2.5.3 Exercise 3.3: controllability

The discrete double integrator

$$x_1[n+1] = x_1[n] + x_2[n], \qquad x_2[n+1] = x_2[n] + u[n], \qquad x_1, x_2, u \in \mathbb{Z}$$

maps $s_{i,j}$ to $(i + j, j + u)$. Figure 2.7 shows the edges for $u = -1, 0, +1$. From $s_{0,0}$, one step gives $(0, u_0)$, so $x_1 = 0$ and no state with $i > 0$ is reachable. Two steps give $(u_0, u_0 + u_1)$; choosing $u_0 = i$ and $u_1 = j - i$ reaches any $s_{i,j}$. For an arbitrary start s_{i_1,j_1} and goal s_{i_2,j_2}, the two-step controls are

$$u_0 = i_2 - i_1 - 2j_1, \qquad u_1 = j_2 + j_1 + i_1 - i_2.$$

One step is not enough in general because it fixes $x_1[1] = i_1 + j_1$.

For continuous LTI systems, the four pairs are tested with the controllability matrix. $(\mathbf{A}_1, \mathbf{B}_1)$ and $(\mathbf{A}_2, \mathbf{B}_2)$ have rank one and are not controllable: in the first, x_1 is decoupled from the input; in the second, $z = x_1 - x_2$ satisfies $\dot{z} = z$ independently of u. $(\mathbf{A}_3, \mathbf{B}_3)$ and $(\mathbf{A}_4, \mathbf{B}_4)$ are controllable. For $(\mathbf{A}_4, \mathbf{B}_4)$ the controllability matrix has rank four; the eigenvalues are $0, 0, \pm\sqrt{2}$ and the PBH test is full rank at each eigenvalue. In the second-order interpretation, $(\mathbf{A}_3, \mathbf{B}_3)$ is fully actuated (one degree of freedom, one input), while $(\mathbf{A}_4, \mathbf{B}_4)$ is underactuated ($q = [x, \theta]$, only x actuated). This again separates controllability from full actuation.

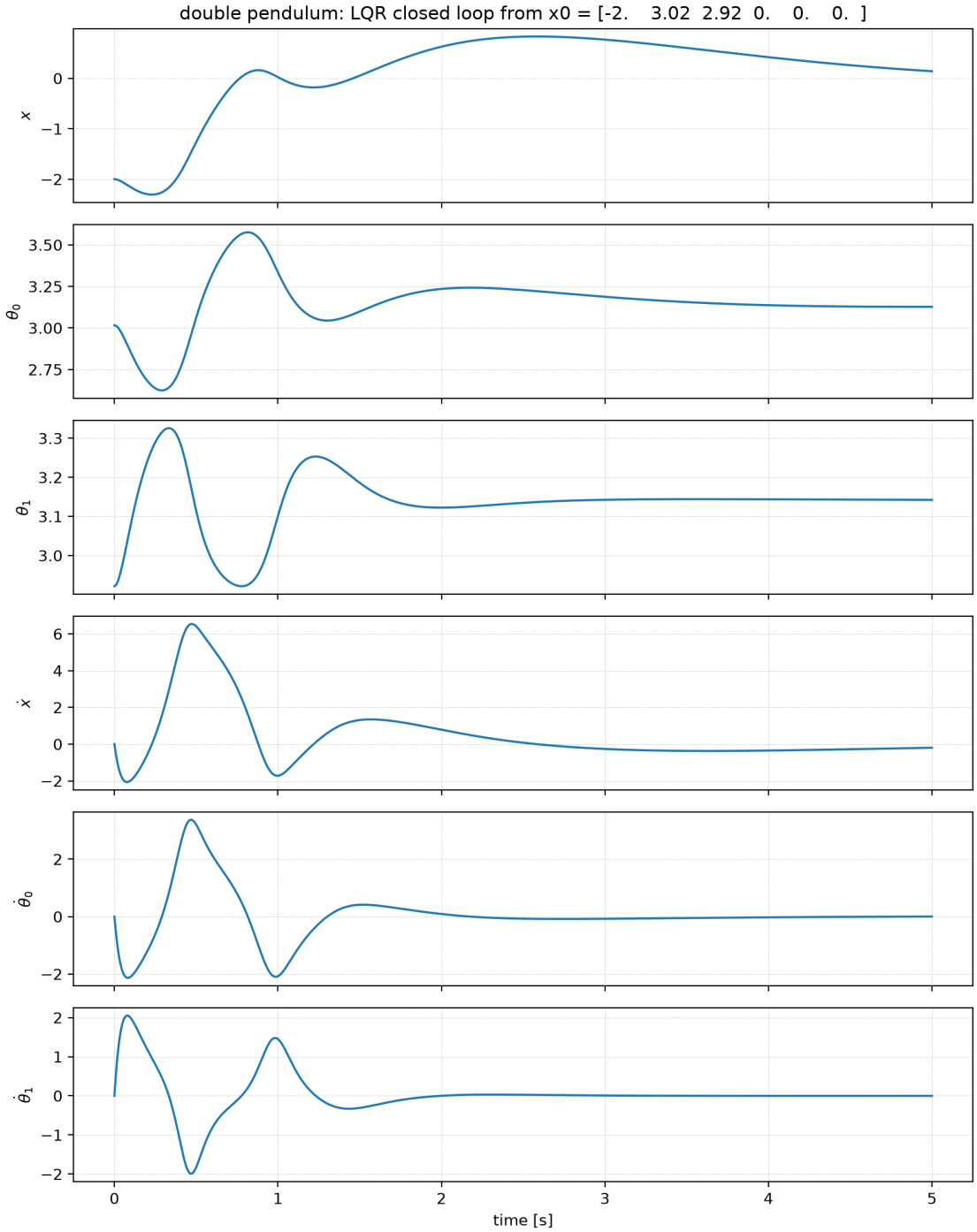


Figure 2.6: Double-pendulum cart-pole under LQR. The angle errors at 5 s are 0.014 and 0.0007 rad, but the cart is still moving; the slow pair of poles has a time constant of about 1.3 s.

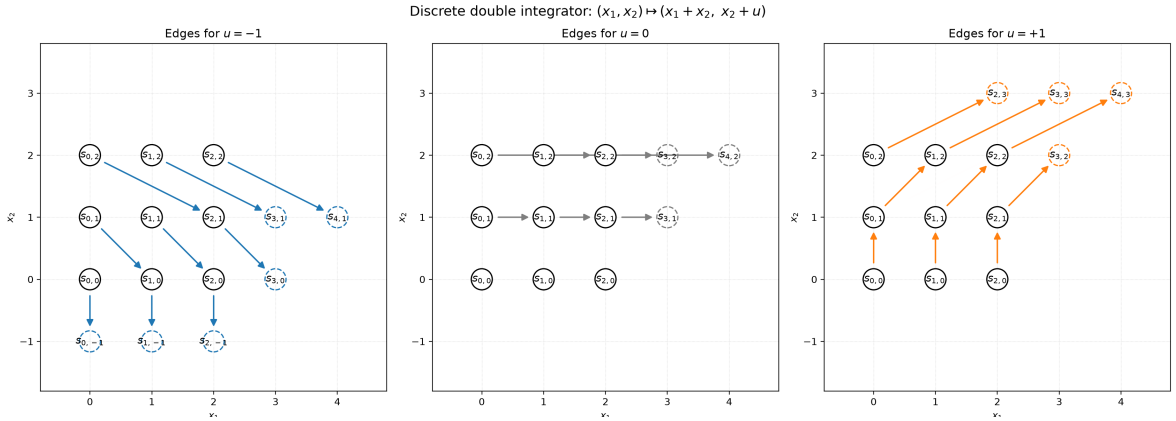


Figure 2.7: Edges of the discrete double integrator for $u = -1, 0, +1$. The source nodes are solid; destinations outside the 3×3 grid are dashed. The two-step construction $s_{0,0} \to (0, i) \to (i, j)$ is the path used in the text.

2.5.4 Exercise 3.4: stability and linearisation

For the pendulum with $m = l = 1$ and damping b,

$$\dot{\theta} = \omega, \qquad \dot{\omega} = -g \sin\theta - b\omega,$$

the equilibrium points are $(0, 0)$ and $(\pi, 0)$, with Jacobian

$$A(\theta^\star) = \begin{bmatrix} 0 & 1 \\ -g\cos\theta^\star & -b \end{bmatrix}.$$

The eigenvalues are $\pm 3.1321j$ for $b = 0$ at the bottom, ± 3.1321 for $b = 0$ at the top, $-0.5 \pm 3.0919j$ for $b = 1$ at the bottom, and $2.6718, -3.6718$ for $b = 1$ at the top.

The stability table follows from these values and from the phase portraits. At the undamped bottom the nonlinear system has closed orbits: it is stable i.s.L. but not asymptotically or exponentially stable. The linearisation is a centre; when the real part is zero the linearisation theorem is inconclusive, and the energy argument is needed. At the top the saddle structure makes both the nonlinear system and its linearisation unstable. With damping, the bottom becomes locally exponentially stable; the top still has one positive eigenvalue, so damping alone cannot stabilise the inverted pendulum.

2.6 Implementation notes

I ran the three example notebooks and the two exercise notebooks in the environment described in the introduction. The Monte-Carlo studies use fixed seeds and 100 trials; the force and torque limits are stated with each table. The acrobot and cart-pole parameters are the defaults from Drake and the `underactuated` package. For the quadrotor, the planar notebook was run headlessly and the 3D notebook uses the official stabilising controller; the sensitivity numbers come from `quadrotor_analysis.py`. The supplied controller equations were not changed; the additional code is logging and the loop over random initial conditions.

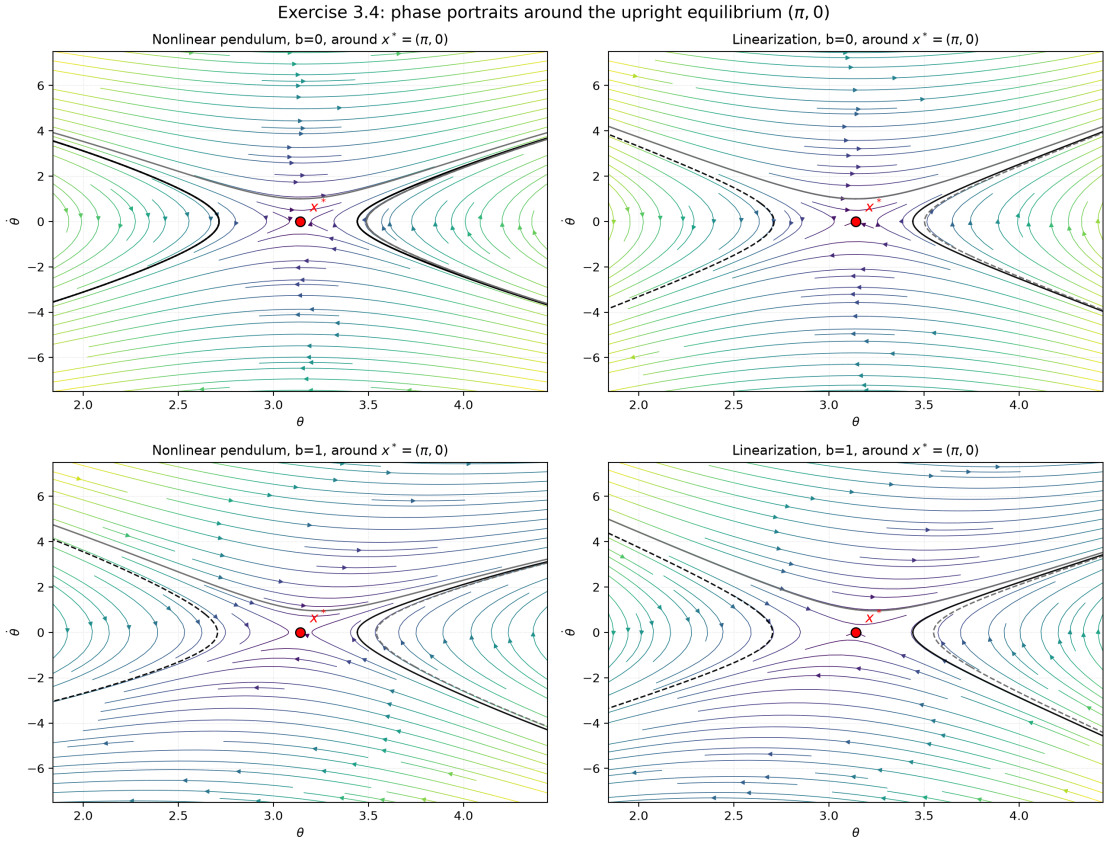


Figure 2.8: Phase portraits around the upright equilibrium for $b = 0$ (top) and $b = 1$ (bottom). The stable manifold separates initial conditions that fall left from those that fall right; damping does not remove the saddle.

2.7 Discussion

The three examples share a workflow, but they fail in different ways. The acrobot and cart-pole are limited by saturation and by the non-minimum-phase undershoot; the quadrotor is limited by the absence of integral action under a constant wind. The Section 3.8 exercises show that the linear model is local: at $\theta = \pi/2$ the linearisation error is already larger than the field itself. A natural next step would be a swing-up controller for the acrobot and cart-pole and an integral or MPC layer for the cart and quadrotor. These are outside the scope of this assignment, but they explain why the local LQR results should not be read as global controllers.

Chapter 3

Simple Models of Walking and Running

3.1 Example 4.3: The Rimless Wheel

The rimless wheel has $n = 8$ spokes, no rim, a point mass at the hip, radius $l = 1$ and slope γ [2]. During stance it behaves as a simple pendulum,

$$\ddot{\theta} = \frac{g}{l} \sin\theta,$$

with energy $E = \frac{1}{2} m l^2 \dot{\theta}^2 + mgl\cos\theta$. At each foot strike the new spoke becomes the stance leg and the collision map is

$$\dot{\theta}^+ = \dot{\theta}^- \cos(2\alpha).$$

For $\alpha = \pi/8$ the kinetic energy is multiplied by $1/2$ at every step. The step-to-step return map is

$$\omega_{n+1} = \cos(2\alpha)\sqrt{\omega_n^2 + 4\frac{g}{l}\sin\alpha\sin\gamma}.$$

A step is possible only above $\omega_1 = \sqrt{2(g/l)[1-\cos(\gamma-\alpha)]}$. The rolling fixed point is

$$\omega^\star = \cot(2\alpha)\sqrt{4\frac{g}{l}\sin\alpha\sin\gamma},$$

and its multiplier is $\cos^2(2\alpha) = 1/2$. The fixed point exists only if it lies above the step threshold.

The slope study gives a clear threshold. At $\gamma = 0.02$ and 0.05 the fixed point is below ω_1 and the wheel cannot sustain a rolling gait; at 0.08, 0.12 and 0.20 the fixed point is valid. The multiplier is $1/2$ in all cases, so the stability of the rolling gait is independent of the slope once the fixed point exists. The fixed point exists only where the gravity input per step can replace the kinetic energy lost at impact; below that slope the map has no valid fixed point.

3.2 Example 4.4: The Compass Gait

The compass gait has two point-mass legs of mass $m = 5$ kg and length $l = 1$ m ($a = b = 0.5$ m), and a hip of mass $m_h = 10$ kg. The state is $[\theta_{st}, \theta_{sw}, \dot{\theta}_{st}, \dot{\theta}_{sw}]$. The stance foot is pinned, the swing leg moves freely, and at heel strike the roles of the two legs are swapped. The textbook initial condition is $[0, 0, 0.4, -2.0]^\top$ with zero hip torque and slope $\gamma = 0.0525$ rad.

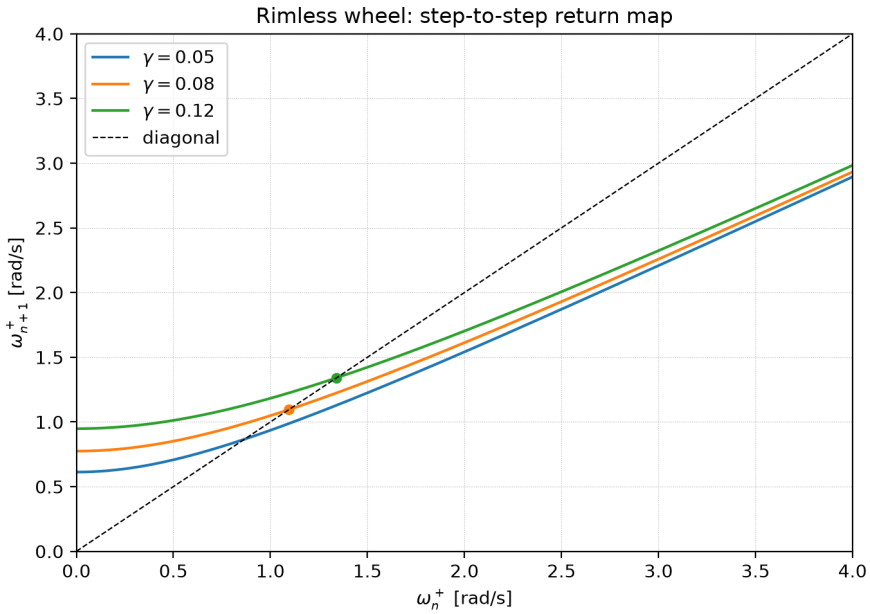


Figure 3.1: Rimless-wheel return map for three slopes. Only slopes above about 0.067 rad have a fixed point above the step threshold; at 0.05 rad the wheel cannot complete a step.

The phase portrait approaches a closed orbit. A step detector based on the plant's "left support" state counts 14 steps in 10 s, with a mean period of 0.735 s. The post-impact states converge quickly to a neighbourhood of the limit cycle. The basin is narrow: in a four-initial-condition test only the nominal state stayed bounded, and in a 3×3 grid of initial states only 3 of 9 runs reached a sustained gait at the shallow slopes; at $\gamma = 0.08$ and 0.12 none of the nine runs did. The passive gait is therefore a valid demonstration, but not a robust walking controller.

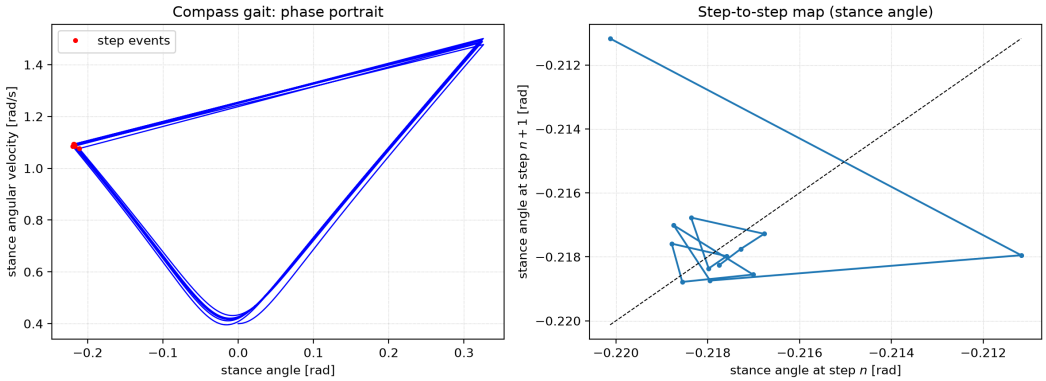


Figure 3.2: Compass gait: phase portrait and step-to-step map. The step points converge to the fixed point after the initial transient; perturbations outside the narrow basin fall.

The contrast with the rimless wheel is instructive. The wheel has many spokes and a collision geometry that naturally returns the state to the rolling solution. The compass gait has only two legs, so the swing-leg landing must match the previous step closely. The energy balance can still produce a limit cycle, but the basin is much smaller.

3.3 Examples 4.5 and 4.6: SLIP and the Apex-to-Apex Map

The spring-loaded inverted pendulum models running with a point mass, a massless spring leg and a touchdown angle that is reset during flight. In flight the dynamics are ballistic; in stance,

$$m\ddot{r} - mr\dot{\theta}^2 + mg\cos\theta - k(r_0 - r) = 0, \qquad mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta} - mgr\sin\theta = 0.$$

The mechanical energy

$$E = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + mgr\cos\theta + \frac{1}{2}k(r_0 - r)^2$$

is conserved because the leg and foot are massless. Unlike the walking models, there is no impact loss; the spring stores and returns the energy.

At the apex the vertical velocity is zero, and energy conservation makes the horizontal velocity a function of the apex height. The return map is therefore one-dimensional, $z[n+1] = P(z[n])$. For the nominal condition $\theta_{td} = 30°$ and $\tilde{E} = 1.61$ the stable fixed point is $z^\star = 0.8776$ with $P' = 0.420$, and the unstable one is $z_u = 0.9163$ with $P' = 1.288$. The textbook values are 0.8768 and 0.9168, with derivatives 0.488 and 1.346; the difference is due to the finite grid and linear interpolation.

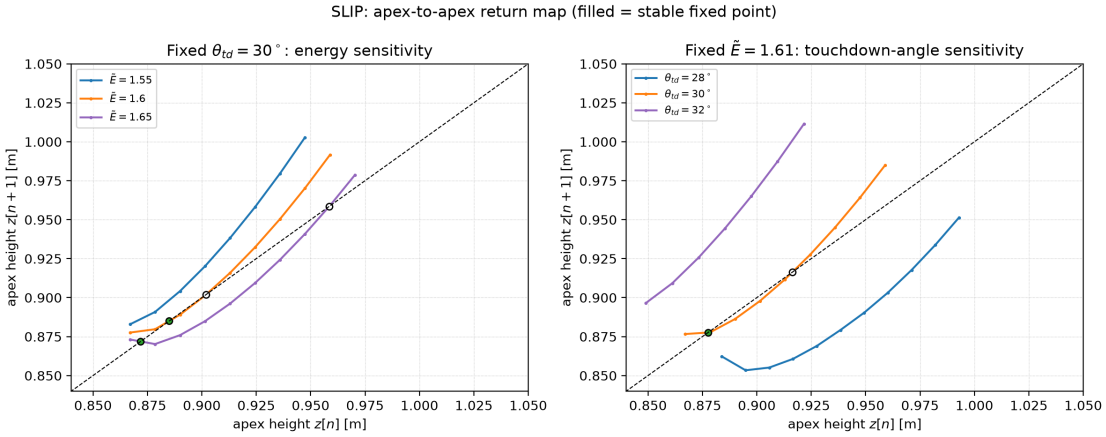


Figure 3.3: SLIP apex-to-apex map. The stable fixed point is at 0.8776 m with $P' = 0.42$; the unstable one is at 0.9163 m with $P' = 1.29$. Filled points are stable fixed points.

The sensitivity study shows a narrow operating window. At 30° a stable fixed point exists for $\tilde{E} = 1.60, 1.61, 1.65$; changing the touchdown angle to 28° or 32° removes the fixed point in the scanned range. The one-dimensional map is obtained on a constant-energy slice, so $|P'| < 1$ proves stability only for perturbations that preserve the energy. Terrain changes and pushes change the energy and are outside the map; this is the partial stability discussed in the notes. The apex map is useful for one-step controllers, but a real robot still needs energy regulation.

3.4 Section 4.6: The One-Dimensional Hopper

The hopper has a body m_b, a leg m_l, a spring k and a preload $\delta \geq 0$. The state is $[z_b, l, \dot{z}_b, \dot{l}]$. The controller changes the preload twice per cycle: at the bottom it applies $\bar{\delta}$, and at the apex it resets the preload to zero. The mechanical energy is

$$E = \frac{1}{2}m_b\dot{z}_b^2 + \frac{1}{2}m_l(\dot{z}_b + \dot{l})^2 + m_b g z_b + m_l g(z_b + l) + \frac{1}{2}k(l + \delta)^2.$$

Equating the energy at the bottom after the preload, at lift-off after the inelastic impact, and at the apex gives

$$\bar{\delta} = \frac{1}{2} z_b + \frac{m_b g}{k} - \frac{(m_b + m_l)^2}{k m_b z_b} g h.$$

If this value is negative the system already has too much energy, and the controller uses $\bar{\delta} = 0$.

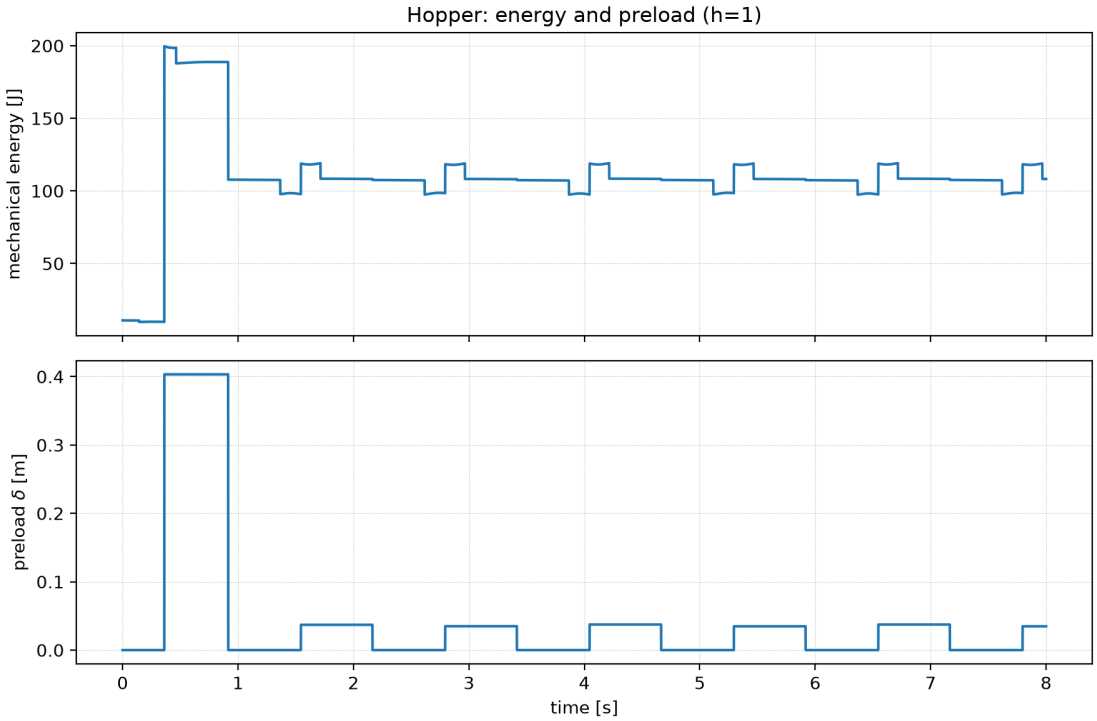


Figure 3.4: Hopper energy and preload for $h = 1$ m. The two energy drops are the touchdown and lift-off impacts; the preload pulses at the bottom restore the energy.

The controller is implemented as a `LeafSystem` that detects the bottom and apex by a sign change of $\dot{z}_b$ and sets the preload accordingly. The autograder gives 13/13. In an 8 s simulation the final-two-second maximum height is within 6 mm of the target for $h = 0.5, 0.75, 1.0, 1.25, 1.5$ m. The maximum preload grows from about 0.024 to 0.302 m over this range, as expected from the energy balance. The controller is also insensitive to the initial body height: starting from $z_b(0) = 0.1$ to 1.0 m, the steady hopping height is the same.

3.5 Implementation notes

I used the supplied rimless-wheel, compass-gait and SLIP notebooks and implemented the hopper controller. Steps in the compass gait are detected from the plant's "left support" state; the rimless return map is computed from the formula in the notebook; the SLIP map uses the notebook's `SpringLoadedInvertedPendulum`. The hopper controller computes the preload at the bottom and resets it at the apex; the autograder gives 13/13. For the hopper sensitivity, each run is 8 s and the last 2 s are used for the height statistics.

3.6 Discussion

The passive walkers and SLIP illustrate two different stability mechanisms. The rimless wheel and the compass gait rely on impact dissipation, and their basins are set by the collision geometry; the compass gait's basin is much narrower because it has only two legs. SLIP has no impact loss, so the apex map is conservative and the stability is only partial. The hopper adds a controller that restores the lost energy. A limitation of these models is the idealisation: point masses, no foot slip, no ground compliance, and, for the hopper, one-dimensional motion. A real robot would need a more detailed contact model and a controller that handles terrain variation.

Chapter 4

Highly-articulated Legged Robots

4.1 Example 5.1: The ZMP Planner

The first example is the ZMP planner for a humanoid [3]. The model is the linear inverted pendulum, with the centre of mass at height z_c and the zero-moment point (ZMP) as the control point. In the notebook the ZMP is tracked by a linear controller; the nominal plan is a sequence of foot placements and a reference ZMP. The CoP identity is checked numerically and holds to machine precision. With the default weights the ZMP tracking error is about 0.153 m; after a velocity push, the feedback policy reduces the final error to 1.5×10^{-5} m. The timing sensitivity study shows that the tracking error grows when the step timing is shortened, because the LIPM has less time to move the CoM between supports.

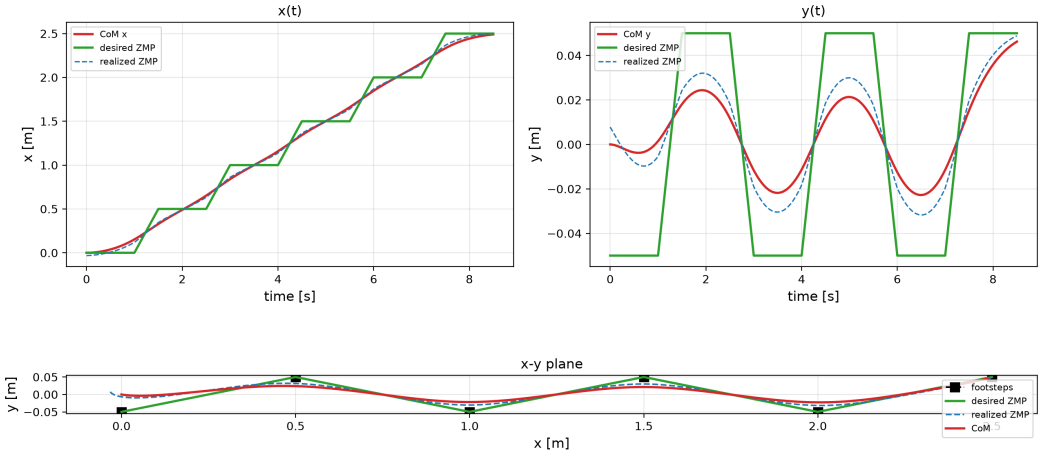


Figure 4.1: ZMP planner: CoM and ZMP trajectories for the nominal footstep plan. The default weights give a ZMP tracking error of 0.153 m; after a velocity push the feedback run returns to 1.5×10^{-5} m.

4.2 Example 5.2: LittleDog Gait Optimization

The second example optimises a walking gait for the LittleDog model. The decision variables are the gait timing and the contact schedule; the constraints come from the centroidal dynamics and the foot geometry. The notebook uses SNOPT. In the executed version the optimisation converges and gives a stride period of about 0.6875 s and a CoM height between 0.131 and 0.148 m. The result is a periodic gait with alternating support phases. The gait is therefore an optimisation problem over contacts; the feedback controller only tracks the resulting trajectory.

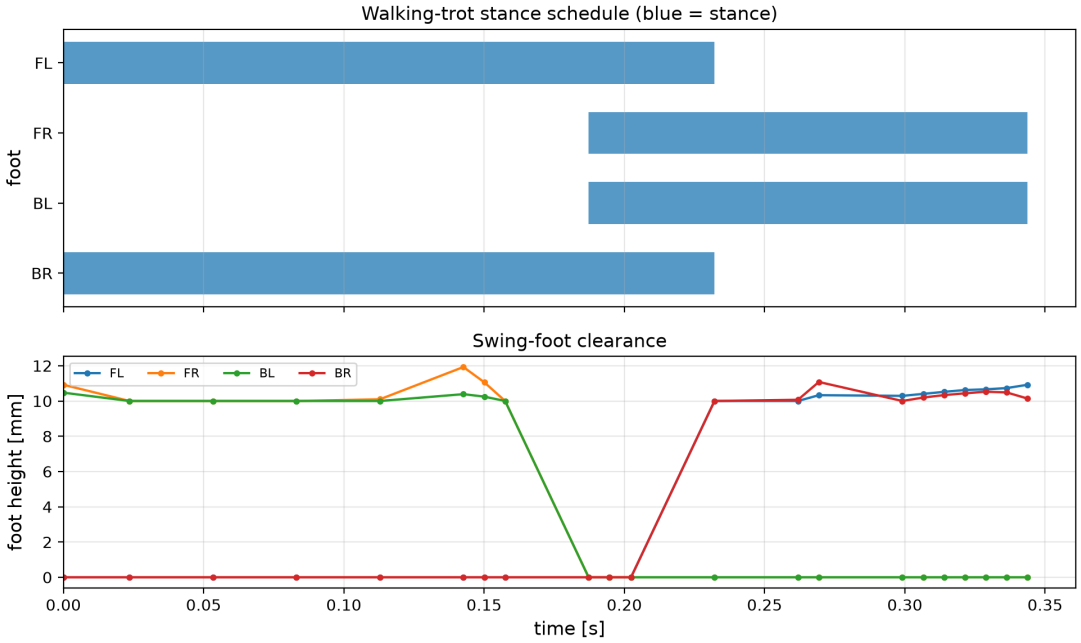


Figure 4.2: LittleDog gait: optimized contact schedule and body trajectory. The stride is 0.6875 s and the CoM height varies between 0.131 and 0.148 m.

4.3 Section 5.8: Exercises 5.1–5.3

The section contains three exercises.

4.3.1 Exercise 5.1: Footstep Planning via MIQP

The terrain and the footstep sequence are encoded with binary variables. The constraints enforce that each foot is assigned to one stepping stone, that the relative foot positions are feasible, and that the chosen footholds are inside the terrain. The objective is the sum of the step lengths. On the two benchmark terrains the MIQP finds costs 4.1593 and 5.6721 and passes the autograder 16/16.

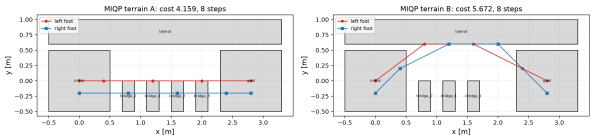
4.3.2 Exercise 5.2: Understanding the Zero-Moment Point

This is an analytic exercise. It derives the ZMP for a free-body diagram, first without an ankle actuator and then with an ideal ankle actuator, and uses the result to discuss foot design. The ZMP is determined by the contact wrench and the CoM acceleration, not simply by the centre of pressure;

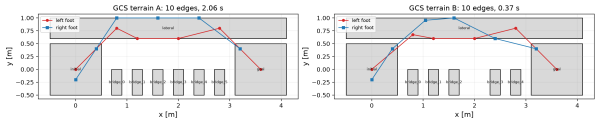
the ankle torque shifts it inside the support polygon. The exercise also shows the range of motion of the actuated ankle for a given CoM trajectory (about 76° to 99° in the numerical example).

4.3.3 Exercise 5.3: Footstep Planning via GCS

The GCS formulation represents the terrain as a graph of convex sets. The shortest path in the graph is found with a mixed-integer or convex optimisation, and the result is a sequence of convex regions. The GCS solution uses 10-edge paths and passes the autograder 16/16. Compared with the MIQP, the GCS formulation is more structured and scales better when the terrain can be decomposed into convex pieces; the MIQP is more direct for small problems.



(a) MIQP footstep plan.



(b) GCS footstep plan.

Figure 4.3: Section 5.8 footstep plans on the benchmark terrains. The MIQP costs are 4.159 and 5.672 for terrains A and B; the GCS plans use 10 edges.

4.4 Implementation notes

I ran the ZMP and LittleDog notebooks and the two footstep-planning notebooks. The LittleDog notebook's default SNOPT iteration limit was too small for the gait to converge, so the major iteration limit was raised to 100; the reported stride is 0.6875 s. The MIQP and GCS exercises use the benchmark terrains in the notebooks and pass 16/16 each. The ZMP exercise is analytic; the numerical values come from `zmp_exercise_analysis.py`.

4.5 Discussion

The ZMP planner, the LittleDog optimisation and the footstep planners address different levels of the same problem. The ZMP planner gives a CoM reference but assumes a fixed contact sequence; the LittleDog optimisation chooses the timing and contacts for a periodic gait; MIQP and GCS choose discrete footholds. The MIQP is direct but scales poorly with the number of steps; GCS uses a convex decomposition and scales better, at the cost of a more structured model. The costs are sensitive to the solver settings and to the terrain discretisation, so they should be read as benchmark values rather than universal performance numbers.

Part II

Manipulation

Chapter 5

Introduction to Manipulation and Drake

5.1 Examples 1.1 and 1.2: Teleoperation in 2D and 3D

The Manipulation introduction starts with teleoperation [4]. The notebook builds a HardwareStation with an IIWA14 and a WSG gripper, adds a differential inverse-kinematics block and connects a set of pose sliders. The data flow is simple: the sliders provide a desired end-effector pose, differential IK converts it into joint positions, the station's position controller tracks the joint command, and the measured pose is fed back to the sliders.

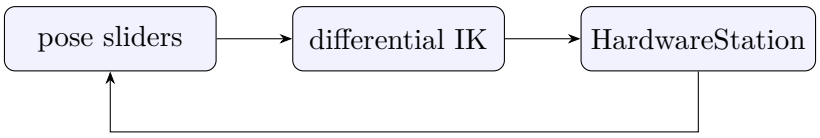


Figure 5.1: Teleoperation data flow. The sliders command a full pose, diff-IK converts it to joint positions, and the measured pose is fed back.

5.1.1 Background: Example 1.1 – Teleoperation in 2D

The 2D example restricts the visible task to roll, x and z; the planar IIWA14 has three actuated joints, so the task and the arm are matched in the nominal configuration.

5.1.2 Example 1.2: Teleoperation in 3D

The 3D example uses the full 7-DoF IIWA14 and a 6-DoF pose slider. The arm is now redundant: many joint configurations produce the same end-effector pose, and differential IK chooses the one closest to the current configuration. In both cases the sliders and the gripper button are the operator interface; the controller and the plant are hidden behind the station ports.

The differential IK solves a local least-squares problem

$$\min_{\dot{q}} \ \|J(q)\dot{q} - \dot{X}_d\|^2 + \lambda\|\dot{q}\|^2$$

subject to joint limits and integrates $\dot{q}$. It is fast and degrades gracefully: when a target is outside the workspace or near a singularity, the solver reports `stuck` instead of producing a discontinuous

jump. The 3D demo also shows the limitation of purely kinematic teleoperation: the clutter is not avoided, because obstacle avoidance is a separate layer.

5.2 Example 1.3: System Diagrams

The System Diagrams example makes the modelling paradigm explicit. A manipulation system is a `Diagram` of smaller systems, each with its own state and ports. The teleoperation demo is already such a diagram: a plant, a controller, a visualizer, and a slider interface. The interactive example lets the reader expand the `iiwa_controller` block and see the hidden complexity inside a single box.

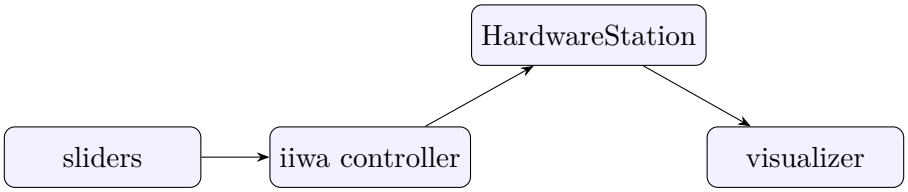


Figure 5.2: A system diagram in the spirit of Example 1.3. The station is a nested diagram; expanding the IIWA controller shows the position controller and model driver inside.

The benefit of the block-diagram approach is composability: a controller, filter or plant can be replaced without rewriting the rest of the system. The cost is that state and ports must be declared explicitly, which can feel like overhead for a small script. For a manipulation stack with perception, planning, control and simulation, the structure is what keeps the parts interchangeable. The HardwareStation scenario used in Exercise 1.4 is a declarative description of the same idea.

5.3 Section 1.7: Exercises 1.1–1.4

Exercise 1.1: Familiarize yourself with Drake. The three documentation questions have short answers:

1. In the `dynamical_systems` tutorial [5] the initial condition of `SimpleContinuousTimeSystem` is set to $x(0) = 0.9$.
2. In the C++ doxygen page on spatial vectors, angular acceleration is written `alpha`; angular velocity is `w`, torque is `t`, and the translational quantities are `v`, `a` and `f`.
3. The Python fitted-value-iteration unit test sets `options.convergence_tol = 1.0`. The default value of `DynamicProgrammingOptions` is `1e-4`, but the test deliberately uses the larger tolerance.

Exercise 1.2: Drake Systems Fundamentals. The exercise implements an inverted pendulum as a `LeafSystem` with state $[\theta, \dot{\theta}]$, a torque input and an angle output. The dynamics are $\ddot{\theta} = -g \sin\theta + \tau/(ml^2)$; for the default $m = l = 1$ this is $-9.81 \sin\theta + \tau$. The system is wired into a diagram with a constant torque source and a logger, and simulated with Drake's `Simulator`. The two verification cases give $\theta(2.5) = 0.0052$ rad and $\theta(1.8) = -0.1074$ rad. The autograder gives 70/70.

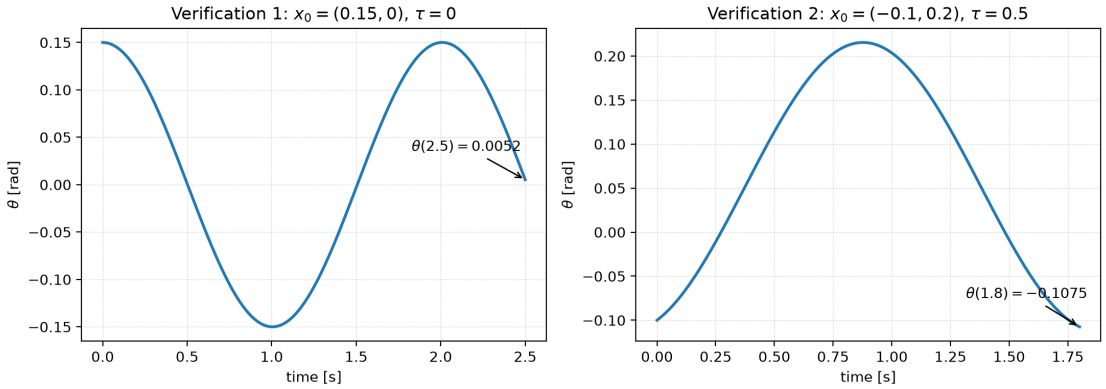


Figure 5.3: Inverted-pendulum simulations for the two verification conditions. The final angles are 0.0052 and −0.1074 rad, matching the autograder.

Exercise 1.3: Physics Simulation and Robot Loading. The exercise loads a 7-DoF IIWA14 with `Parser` and `MultibodyPlant`, welds the base, and adds a simple proportional controller

$$\tau = -K(q - q_d)$$

with 14 inputs (seven positions and seven velocities) and seven torque outputs. It then builds a table from an SDF string, generates letter assets with the provided API, and assembles a full scene. The visible autograders give 10/10 for the IIWA14 structure, 10/10 for the controller, 10/10 for the full-system run, and 20/20 for the Gradescope verification, for a total of 50/50. The verification target is

$$q_{\text{final}} = [0.0126, 0.0081, -0.0573, -0.0244, -0.0001, 0.0005, 0.0000].$$

The small residuals are due to gravity and joint damping; the P controller does not compensate gravity.

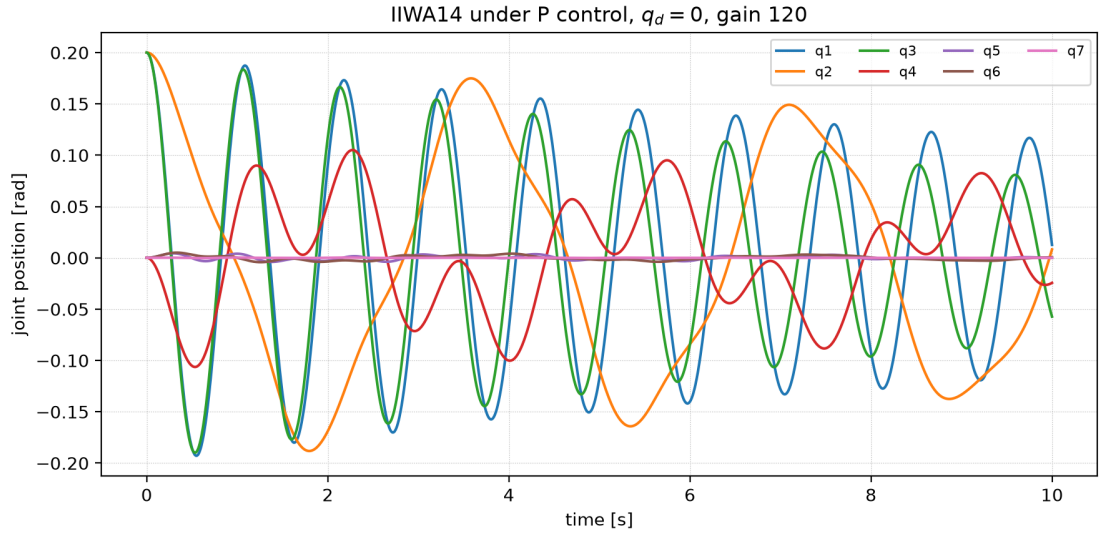


Figure 5.4: Joint positions of the IIWA14 under the proportional controller. The residuals are below 0.06 rad; the P controller does not compensate gravity, so a small steady-state offset remains.

Exercise 1.4: HardwareStation and Advanced Scenarios. This exercise replaces the manual wiring with a HardwareStation and a YAML scenario. The scenario contains two IIWA14 robots, a local table, three letters and a free YCB object. The station is added to a diagram and simulated. The autograders give 10/10 for the basic station, 10/10 for the bimanual station, 15/15 for the full-system structure with eight model instances, and 15/15 for the simulation run, for a total of 50/50.

Listing 5.1: Scenario directives for the bimanual station (abridged).

```
directives:
- add_model:
    name: iiwa_left
    file: package://drake_models/iiwa_description/urdf/iiwa14_primitive_collision.urdf
- add_weld:
    parent: world
    child: iiwa_left::iiwa_link_0
- add_model:
    name: iiwa_right
    file: package://drake_models/iiwa_description/urdf/iiwa14_primitive_collision.urdf
- add_weld:
    parent: world
    child: iiwa_right::iiwa_link_0
    X_PC:
        translation: [0.0, 1.0, 0.0]
- add_model:
    name: table
    file: file:///.../assets/table.sdf
- add_weld:
    parent: world
    child: table::table_link
    X_PC:
        translation: [0.5, 0.0, -0.05]
```

5.4 Implementation notes

I used the pinned `drake_models` package for the IIWA14, the WSG gripper and the YCB object. The package was not present in the initial environment; after it was placed in the local Drake cache, all notebooks ran with `package://drake_models`. The teleoperation notebook was run with `COLAB_TESTING=1`; the 2D and 3D functions were simulated for 0.1 s. The exercise notebooks use the official model and the local table/letter assets, and the autograders give 70/70, 50/50 and 50/50.

5.5 Discussion

The teleoperation stack is deliberately local: differential IK tracks a pose, but it does not avoid obstacles and it can get stuck near singularities. The System Diagrams example explains why the stack is organised as a diagram rather than a single script; the HardwareStation scenario makes the same structure declarative. The P controller in Exercise 1.3 is also intentionally simple and leaves a gravity-induced steady-state error. For a real manipulation system, the next steps would be a trajectory planner, a collision-aware IK or MPC layer, and a gravity-compensating controller.

Chapter 6

Conclusions

The four chapters share a common methodological pattern. A model is written in terms of state and inputs; a local linearisation is used to design a controller; the nonlinear simulation is used to find the limits of the linear design. The limits appear in different forms in each chapter: saturation and a finite basin for the acrobot and cart-pole; impact losses and a narrow basin for the compass gait; energy conservation and partial stability for SLIP; and contact planning for the legged robots.

The most consistent lesson is that underactuation and controllability answer different questions. The acrobot, cart-pole and quadrotor are all underactuated, yet their linearisations are controllable. What changes from one system to the next is how much input authority is available in each direction, and the singular values and nonlinear simulations show that this difference can be the difference between a comfortable recovery and a saturated failure.

A second observation is that LQR is a local tool. The closed-loop eigenvalues give local exponential stability, but they say little about the basin of attraction or about what happens when the actuator saturates. The acrobot and cart-pole runs make this concrete: the same controller that recovers from a small perturbation fails from a larger one or from a different direction.

The hybrid systems in Chapter 4 require a different object. The limit cycle and its return map replace the fixed point and the Jacobian. Impact maps are dissipative and can produce a stable gait, but the basin can be narrow; a conservative model such as SLIP gives a one-dimensional apex map and only partial stability. The hopper controller is a good example of using the energy balance directly instead of trying to track a trajectory.

Finally, planning and control are complementary. The ZMP planner, the LittleDog optimisation, MIQP and GCS all choose a contact sequence or a reference trajectory; the feedback controller only tracks it. The software stack—Drake diagrams, URDF/SDF assets, differential IK and HardwareStation—is what makes it possible to move between these levels without rewriting the whole system.

Bibliography

[1] R. Tedrake, *Underactuated Robotics*, Chapter 3: Acrobots, Cart-Poles, and Quadrotors. https://underactuated.csail.mit.edu/acrobot.html

[2] R. Tedrake, *Underactuated Robotics*, Chapter 4: Simple Models of Walking and Running. https://underactuated.csail.mit.edu/simple_legs.html

[3] R. Tedrake, *Underactuated Robotics*, Chapter 5: Highly-articulated Legged Robots. https://underactuated.csail.mit.edu/humanoids.html

[4] R. Tedrake, *Robotic Manipulation*, Chapter 1: Introduction. https://manipulation.csail.mit.edu/intro.html

[5] Drake tutorials, including the dynamical systems and authoring leaf systems tutorials, and the C++ doxygen page on spatial vectors. https://github.com/RobotLocomotion/drake/tree/master/tutorials, https://drake.mit.edu/doxygen_cxx/group__multibody__spatial__vectors.html

[6] Course lectures: 6.8210 Lecture 5 (Acrobots, cart-poles and quadrotors), 6.832 Lecture 15 (Simple models of walking), 6.8210 Lecture 16 (Humanoid robots), and 6.4210 Lecture 1 (Introduction to manipulation).

Appendix A

Code and Reproduction

The code is organised by chapter:

Chapter	Main files
Ch. 3	`Ch-3/Ch3_Assignment_Report/code/`
Ch. 4	`Ch-4/Ch4_Assignment_Report/code/`
Ch. 5	`Ch-5/Ch5_Assignment_Report/code/`
Ch. 1	`Ch-1/Ch1_Assignment_Report/code/`

The notebooks can be executed headlessly with

```
COLAB_TESTING=1 /Users/leo/venvs/drake/bin/jupyter nbconvert \
  --to notebook --execute --inplace --ExecutePreprocessor.timeout=300 \
  <notebook>.ipynb
```

and the reports are compiled with `pdflatex` run twice. The combined code is in `Assignment_1_Code.zip`.

A.1 Selected code

Cart-pole dynamics and LQR.

```
def f(x, u):
    c, s = np.cos(x[1]), np.sin(x[1])
    g = 9.81
    return np.array([
        x[2], x[3],
        (u[0] + s * (x[3]**2 + g * c)) / (1 + s**2),
        (-u[0] * c - x[3]**2 * c * s - 2 * g * s) / (1 + s**2)])
```

Double-pendulum URDF.

```
<link name="pendulum1">
  <inertial>
    <origin xyz="0 0 1" />
    <mass value="1" />
    <inertia ixx="0" ixy="0" ixz="0" iyy="0" iyz="0" izz="0" />
  </inertial>
</link>
<joint name="theta1" type="continuous">
```

```
  <parent link="pendulum0" />
  <child link="pendulum1" />
  <origin xyz="0 0 -1" />
  <axis xyz="0 -1 0" />
</joint>
```

Hopper controller.

```
def get_delta_bar(self, state):
    k, mb, ml, g = self.hopper_parameters
    zb = state[0]
    return (0.5 * zb + mb * g / k
            - ((mb + ml) ** 2) * g * self.h / (k * mb * zb))
```

Cart-pole LQR wiring (Chapter 3).

```
def cartpole_balancing(cartpole_urdf, x_star, Q, R):
    builder = DiagramBuilder()
    plant, scene_graph = AddMultibodyPlantSceneGraph(builder, time_step=0.0)
    Parser(plant).AddModelsFromString(cartpole_urdf, "urdf")
    plant.Finalize()
    plant.set_name("cartpole")

    context = plant.CreateDefaultContext()
    context.get_mutable_continuous_state_vector().SetFromVector(x_star)
    plant.get_actuation_input_port().FixValue(context, [0])
    input_i = plant.get_actuation_input_port().get_index()

    lqr = LinearQuadraticRegulator(
        plant, context, Q, R, input_port_index=input_i)
    lqr = builder.AddSystem(lqr)
    builder.Connect(plant.get_state_output_port(), lqr.get_input_port(0))
    builder.Connect(lqr.get_output_port(0),
                    plant.get_actuation_input_port())
    return builder.Build()
```

Inverted pendulum system (Chapter 1).

```
class InvertedPendulum(LeafSystem):
    def __init__(self, mass=1.0, length=1.0, gravity=9.81):
        LeafSystem.__init__(self)
        self.mass, self.length, self.gravity = mass, length, gravity
        self.DeclareContinuousState(2)
        self.torque_port = self.DeclareVectorInputPort("torque", 1)
        self.DeclareVectorOutputPort("theta", 1, self.OutputTheta)

    def DoCalcTimeDerivatives(self, context, derivatives):
        x = context.get_continuous_state_vector()
        theta, theta_dot = x[0], x[1]
        torque = self.torque_port.Eval(context)[0]
        theta_ddot = -self.gravity * np.sin(theta) + torque / (
            self.mass * self.length**2)
        derivatives.get_mutable_vector().SetFromVector(
            [theta_dot, theta_ddot])
```

```
    def OutputTheta(self, context, output):
        output.SetFromVector([context.get_continuous_state_vector()[0]])
```

Hopper controller logic (Chapter 4).

```
def is_bottom_or_apex(self, state):
    if np.any(np.isnan(self.last_state)):
        return None
    zb, l, zb_dot, l_dot = state
    last_zb_dot = self.last_state[2]
    if last_zb_dot <= 0.0 and zb_dot > 0.0:
        return "bottom" if zb < 0.0 else "apex"
    if last_zb_dot >= 0.0 and zb_dot < 0.0:
        return "apex" if zb > 0.0 else "bottom"
    return None

def DoCalcVectorOutput(self, context, delta):
    state = self.get_input_port(0).Eval(context)
    event = self.is_bottom_or_apex(state)
    if event == "bottom":
        delta.SetFromVector([max(self.get_delta_bar(state), 0.0)])
    elif event == "apex":
        delta.SetFromVector([0.0])
    else:
        delta.SetFromVector([self.last_delta])
    self.last_delta = delta[0]
    self.last_state = np.array(state)
```

HardwareStation scenario. The scenario in Exercise 1.4 uses `add_model`, `add_weld` and `default_free_body_pose` directives; the local table and letters are referenced by absolute `file://` URIs, while the robots and the YCB object use `package://drake_models`.